

Advancing Autonomous UAV Target Localization in GPS-Denied Environments

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Abstract — This research introduces a novel approach to enhance the robustness of autonomous UAV target localization in GPS-denied environments. Typically, localization in such environments relies on utilizing received signal strength (RSS) transmitted by the target and employing the Q-learning algorithm to determine agent actions. Usually, there are two approaches to defining Q-learning's state, both using only a single variable as the Q-learning's state. One of those approaches is potentially hard to converge, while the other is susceptible to unrealistic assumptions in a real-world deployment. This study proposes an improved Q-learning algorithm by incorporating multiple variables as states, addressing the limitations of existing approaches that use a single variable. In this research, it will be proven that if the assumptions don't hold, the performance of the proposed method by another research will be messed up. In addition, it's proved that a new approach is proposed in this research able to outperform the approach using only single-variable as Q-learning state in terms of speeds of convergence and robustness, even when the assumptions are not met.

I. INTRODUCTION

In recent years, unmanned aerial vehicles (UAVs) have grown significantly. The global UAV market is valued at over \$7.7 billion and is projected to reach \$17.5 billion by 2028, with a compound annual growth rate of 14.7% from 2022 to 2028 [1]. The popularity of UAVs is due to their simple flight control systems, making them adaptable to a wide range of applications. One such application is search and rescue (SAR) operations, which are crucial for assisting people in danger or distress, particularly during natural disasters.

The SAR operation is critical so that the rescue can be done immediately. The problem with search operations is that they can occur in environments inaccessible to humans. This is where the role of the UAVs is needed, to help the interaction between the environment and the rescue team, which helps the rescue team localize the target or the victim. However, another problem that often occurs is the need for knowledge of the rescue team in the environment in which natural disasters occur. There are also some situations where the GPS is inaccessible for the rescue team. Those facts motivate many researchers to research the path planner algorithms for UAVs in GPS-denied environments. The path planner algorithm aims

to design the path for the UAV from its starting point to the target location.¹

This paper focuses on radio signal-based techniques for target localization problems using UAVs. Here, the radio signal is emitted from a target's smart device, which can be forced to transmit radio frequency (RF) waves in emergencies [2]. The emitted RF allows them to be detected and tracked by UAVs equipped with an antenna to receive the RF waves. This allows for a more efficient and reliable method of locating the target. This technique is one of the two main techniques used in typical target localization using the UAV, the other one is vision-aided techniques [3], [4]. Using radio signal-based techniques has several benefits over vision-based techniques. Radio signals can penetrate through obstacles such as walls (although they will be attenuated), making them more reliable for detecting signals from survivors under challenging terrains or environments [5].

Research done by [6]–[9] modeled target localization problems based on RF waves as a Markov Decision Process (MDP) problem, which can be solved with a Reinforcement Learning (RL) algorithm. Fig. 1 illustrates the approach taken in research [6]–[9] to localize targets using RF waves. In existing research, it is assumed that the target can transmit RF under certain circumstances so that it is possible to be detected by a UAV equipped with an antenna. Later, the antenna will receive RF waves in the form of received signal strength (RSS), a quantity that states how strong the signal is received by the antenna [2], [6], [8], [10]. RSS is then processed with one of the RL algorithms, namely Q-learning, to determine the direction of motion of the agent [6]–[9].

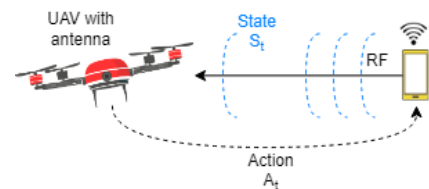


Fig. 1. Application of the RL algorithm for target localization using RF.

Designing Q-learning algorithms to solve target localization problems using RSS is quite difficult. This is due to the propagation characteristics of RF waves. Under ideal

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conditions, RF waves propagate uniformly in all directions so that the RSS received by all points at the same radius should have the same value [11]. This characteristic can cause the Q-learning algorithm to be not robust or difficult to achieve convergence. This is because the agent thinks of several different locations (with the same RSS value) as the same location.

Several approaches in previous research, such as [6], [7] increase the possibility of being unable to reach convergence. This is due to the proposed research's inaccurate Q-learning state definition. While other studies, [8], [9], use unrealistic assumptions, namely that no two or more locations have the same RSS value, contrary to the RF wave propagation characteristics. To overcome this problem, this study proposes a robust target localization algorithm design capable of converging against RF wave propagation characteristics that propagate uniformly to all directions by defining multiple variables as Q-learning's state. In summary, the contribution of this paper can be summarized below:

1. In this research, we discuss potential problems that arise due to the approach taken by previous RF wave-based target localization research.
2. This research offers a new approach as a solution to potential problems arising from approaches taken by research on target localization based on RF, which defines Q-learning's state with multiple variables. The solution proposed has proven can able to converge in dealing with the characteristics of RF wave propagation.

The rest of this paper is organized as follows. In Section II, we discussed the approach taken by other researchers. In Section III, we discuss the background of Q-learning. After that, we analyze the weakness of the approaches that other researchers took and our solution to overcome that problem in Section IV. In Section V, we discuss the simulation setup and analyze the results. Finally, we conclude this research in Section VI.

II. RELATED WORKS

Since the main focus of our research is improving the performance of target localization problems using the Q-learning algorithm, we will only discuss other research that uses Q-learning for their target localization based on the RSS. In typical target localization using RSS based on Q-learning, there are two different approaches how defining the Q-learning state. The first approach involves dividing the UAV search area into grids of a specific size, with each grid represented by a set of RSS values. This approach assumes that no two grids have the same RSS value, which may be an unrealistic assumption [8], [9]. This approach is also not in accordance with the nature of the propagation of RF waves, which propagate uniformly in all directions, which means that all points that have the same distance from the propagation source will have the same RSS value [11]. The second one is the state defined by some intervals of RSS range (e.g., state 1 for RSS value < -50 dBm, state 2 for RSS value between 72 and 50 dBm, etc.) [6], [7]. Worth knowing that those two approaches only use a single variable to define the Q-learning state. Using these two approaches may result in a problem:

representing multiple distinct real-world locations with a single Q-learning state. Those approaches limit the agent to having the same action for multiple locations since a single Q-learning state only represents those multiple locations. This can prevent the agent from finding the optimal policy as various locations may require different actions to reach the target.

TABLE I
LITERATURE REVIEW.

Reference	Method	State Definition
[8], [9]	Q-learning	RSS, the environment is divided into square grids, each 1 x 1 m. Each grid assigned with specific RSS value, and it is assumed that no grid has the same RSS values. The RSS value is obtained from the antenna attached to the UAV.
[6]	Q-learning	RSS, there are eight states, each of which contains an interval of RSS values. The RSS value is obtained from the antenna attached to the UAV.
[7]	Q-learning	RSS, there are ten states, each of which contains an interval of RSS values. The RSS value is obtained from the antenna attached to the UAV.
Our method	Q-learning	RSS, the environment is divided into square grids, each 1 x 1 m. Each grid is assigned the RSS value. Besides that, an additional variable for defining Q-learning's state is proposed, which is the previous action the UAV took.

III. BACKGROUND ON Q-LEARNING

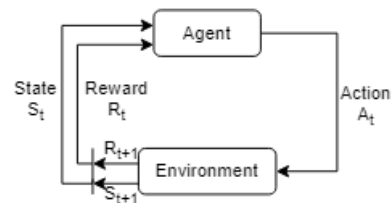


Fig. 2. The agent-environment interaction in the MDP [12].

Target localization problems based on RF waves are usually modelled as Markov Decision Process (MDP), as shown in Fig. 2, a mathematical framework used to model decision-making problems with uncertain outcomes. It can be written as tuple $\langle S, A, P, R, \gamma \rangle$ where $S = \{s_1, \dots, s_I\}$ is the set that contains states, $A = \{a_1, \dots, a_J\}$ is the set that contains actions that may be take by the agent, $R = \{r_t(s_i, a_j) | s_i \in$

$S, a_j \in A$ is set of rewards, $P = \{p(s', s, a_j) | s, s' \in S, a_j \in A\}$ is transition probability, and $\gamma \in (0,1]$ is *discount factor* [13].

At each time step, the agent chooses an action a_t from the set of available actions and moves to a new state s_{t+1} based on the current state s_t and reward r_t . The reward r_{t+1} associated with the transition is then determined. In an MDP, the goal is to find the optimal policy, $\pi^* : s \rightarrow a$, which is the ability to determine the best action based on the current state. In the target-localization problem, the agent, which is the UAV, is in a particular state, and based on its current state, it selects an action, as illustrated in Fig. 2. This action leads the agent to a new state and produces a reward. The agent's goal is to maximize the cumulative reward over time, which will later help the agent find the optimal policy.

One of many algorithms that can be used to solve MDP is Q-learning. Q-learning can be used to find an optimal policy for any MDP. It is a value-based reinforcement learning algorithm that learns the value of an action in a particular state. As the agent interacts with the environment at each time step, the reward r in response to the agent's action a under a given state s will be stored as $Q(s, a)$ or Q-value. The main idea of the algorithm is to build states and actions into a Q-table to store Q-values and select the actions that can get the most benefit according to the Q values. The Q-value is calculated with the Bellman equation as below:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha[r_{t+1} + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t)], \quad (1)$$

where s_t and a_t is the state and the action of the agent at time t , α is the learning rate, r_{t+1} is the reward at time $t+1$ that obtained by the agent after took action at time t . Last, γ is the discount factor determining future rewards' importance [12]. After the iterative process, the agent will eventually learn the optimal Q-values for each state-action pair. The actions with the highest Q-values for each state represent the optimal policy.

Q-learning requires clever exploration mechanisms, as randomly choosing actions without reference to an estimated probability distribution shows poor performance. One method is ϵ -greedy, where $0 < \epsilon < 1$ is a parameter to control the amount of exploration and exploitation made by the agent. With probability $1 - \epsilon$, exploitation enables the agent to choose the action that it has the best long-term effect. Otherwise, with probability ϵ , exploration, the agent chooses the action randomly, intending to maximize the expected reward.

In Q-learning, the Q-values are stored in the table that stores the Q-value of all possible state-action pairs, as shown in Fig. 3. Every interaction between the agent and the environment will update the Q-value based on the observed rewards and the estimated Q-values of the next state. After several training episodes, the agent will receive more information from the environment that will later converge to the true optimal Q-values [14].

		Action	
		a_1	a_n
State	s_1	Qvalue	
	s_n		Qvalue

Fig. 3. Table to store the Q-values.

IV. SYSTEM MODEL

In this section, we will propose the new approach to make the Q-learning algorithm able to converge under the nature of RF waves propagation. Before that, an analysis of potential problems will be carried out in the approach used in existing research. Based on this analysis, there will propose a new approach that can overcome potential problems in the existing research approach.

A. Analysis of Potential Problems on Existing Research Approach

The potential problem that can arise from [6]–[9] is that the proposed algorithm is not robust or unable to converge under the characteristics of RF waves. The ability of an Q-learning algorithm to achieve convergence is related to the ability of the algorithm to obtain the optimal q-value. The optimal q-value defined with the equation below [15].

$$q_*(s, a) \doteq \sum_{s' \in S} P_a(s, s') [r(s, a, s') + \gamma v_*(s')], \quad (2)$$

In equation (2) it can be seen that the optimal Q-value is influenced by $P_a(s, s')$ which is the probability of moving from a state s to a new state s' after taking action a . This shows, assuming other variables are the same, the greater the probability of an agent to move from one state to another, the easier it will be to find the optimal policy. Based on this, it is not enough to define the state variables of Q-learning using RSS alone due to the RF propagation characteristics that have been discussed in Section II. Unfortunately, this state definition occurs in research [6]–[9].

Fig. 4 highlights the significance of assigning a unique state to each location in the real world. The illustration depicts a red target in the center and two blue dots representing possible starting points for the robot on the right and left sides. The objective of Q-learning is to discover the optimal policy, determining the best action to take in each state to maximize cumulative rewards for the agent. The numbers in the squares represent states, and the arrows indicate the fastest path to the target or the optimal policy. In Fig. 4(a), where each state corresponds to a single location, the agent easily reaches the target as the policy guides it along the fastest route. However, in Fig. 4(b), where one state represents multiple locations, such as the square to the right being the same state as the square to the left of the target, the agent lacks location-specific information. Since the learning process aims to identify the optimal policy with a single best action for each state, the agent chooses the same action for both squares, disregarding their

different real-world locations (resulting from the two locations).

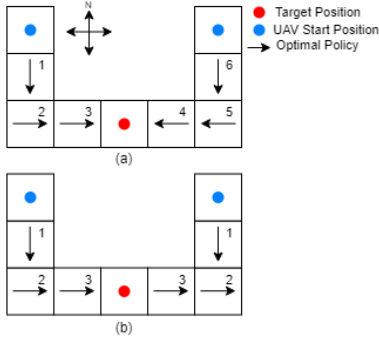


Fig. 4. Illustration of the state-uniqueness importance, (a) Each distinct location represented by a unique state, (b) Multiple distinct locations represented by a single state.

B. Robust Q-learning under RF Waves Propagation

To address the issue mentioned in the section above, this research proposes using two variables as the Q-learning state. The first variable is the RSS value received by the agent. The second variable is the latest action taken by the UAV. This additional state aims to add uniqueness to every distinct area, with the expectation to increase the value of $P_a(s, s')$ in the equation (2) and later improved performance compared to using a single variable to define the Q-learning state.

Since the definitions of Q-learning's rewards, action, and other mechanisms used in research [6]–[9] are varied and the focus of this research is to improve the Q-learning's robustness by defining the Q-learning's state with multiple variables, we will only compare the proposed approach to [8], [9], by defining the rest of Q-learning variable, besides the state, as close to [8], [9]. Nevertheless, the comparison between this research to research [8], [9] can be generalized to research [6], [7] too. This is because the $P_a(s, s')$ part from equation (2) in research [8], [9] higher than [6], [7]. So, if the proposed algorithm has better performance than [8], [9] because successfully produces higher $P_a(s, s')$, it means the proposed algorithm by this research will automatically have a better performance than [6], [7], if we use the state definition from this research and retain the rest of Q-learning variable definition from [6], [7].

Our approach partitions the environment into 1m x 1m grids. The objective of the UAV is to locate the target by moving from the centre of one grid to the centre of an adjacent grid, as illustrated in Fig. 5, at every each time step. A reward is given to the agent for each step it takes. The reward system is defined as follows.

$$R(s, a) = RSS(s') - RSS(s), \quad (3)$$

where $R(s, a)$ represents the rewards of taking action a at state s , $RSS(s')$ represents the RSS value of the new state, and $RSS(s)$ represents the RSS value of the old state. Besides that, the agent will get 100 for successfully reaching the target (at least 1m away from the target) and -10 for not being able to reach the target at the end of the episode, which ends after the

agent takes a maximum of 10000 steps in that episode. This definition of state and reward encourages the agent to move toward the target as quickly as possible while avoiding any obstacles in the grid. Each training episode will end if the agent can reach the target or the maximum number of steps has already been exceeded. Last, the ϵ will start from 1 and will be multiplied with 0.9 after each step, the ϵ will stop decay after its value of 0.001.

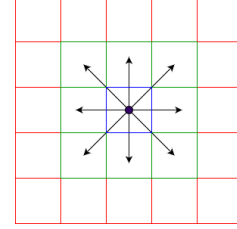


Fig. 5. The agent (in purple) is in the blue state grid. The agent has eight actions to choose from, moving towards the grid that is adjacent to the blue grid (the green grid). Agents cannot move to the red grid.

V. SIMULATION

A. Simulation Setup

Fig. 6 depicts the simulation environment of three rooms measuring 60m x 40m, with obstacles obstructing the UAV's path to the target. Each room has distinct characteristics to assess the algorithm's performance. In Room 1, the agent pursues the target on the left side. Conversely, in Room 2, the agent is positioned on the left, while the target is on the right. Lastly, the target is positioned in the middle of the room, and the room exhibits a symmetric shape. Placing the agent in the middle generates identical RSS values within the same radius, regardless of direction. Room 3 is the hardest environment since it increases the likelihood of multiple locations sharing the same RSS value and it increases the probability agent doesn't know its real location relative to the target.

Each room has an agent (yellow coloured) and a target (green coloured). The environment is divided into a 1x1 grid. Each grid is assigned an RSS value replaced with the A* algorithm. We realize that the A* algorithm will not perfectly simulate RSS propagation compared to the commercially available wireless signal propagation simulator, as our simulated environment assumes perfect RSS attenuation on every wall. However, the focus of this simulation is to study the viability of finding targets using only RSS. So, suppose in this simulation, our method shows better performance than the previous Q-learning. It should remain true in more accurate simulation environments.

In each room, the simulation contains a possible combination of two variables that we want to analyse. The first one is the effect of the assumption that no two grids have the same RSS value, which is not in accordance with the characteristics of RF wave propagation, as assumed in [8], [9]. The environment that follows this assumption will be mentioned as a “unique environment.” Meanwhile, the environment that disobeys that assumption means fulfilled the RF wave propagation characteristics mentioned as a “non-

unique environment.”. The second variable is the effect of adding the last action taken by the agent as a Q-learning state, which is proposed by this research. To produce a “unique environment,” we add random noise to each grid that contains the RSS value.

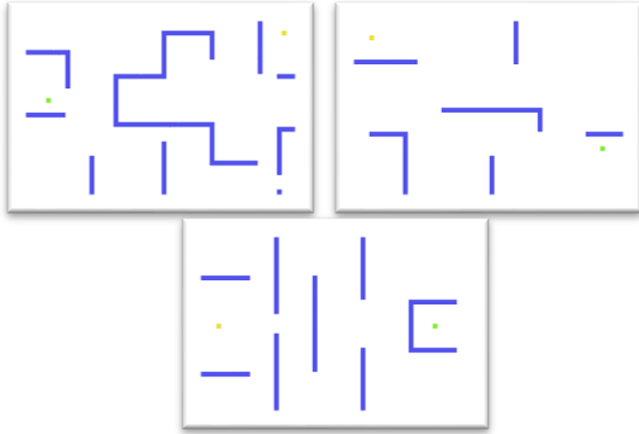


Fig. 6. Simulation environments used to test the algorithms, Room 1 (top left), Room 2 (top right), and Room 3 (bottom).

B. Simulation Results

In this study, the simulation was carried out using the proposed and previous research methods. As mentioned before, the main difference between the two methods is the variables used for the state. Our proposed method is Q-learning which has multiple variables as the state in the form of the RSS value and an additional state in the form of the previous action taken by the UAV. Meanwhile, the previous method is Q-learning with a single variable, sensed RSS value, as the state. Both methods were simulated in two cases. In the first case, no two grids have the same RSS value as assumed in the previous research. In the second case, the RSS value of each grid is not unique.

The simulation results using both methods for UAV in finding a target can be seen in Fig. 7, which depicts the number of steps required to reach the target in each episode. In a unique environment, both algorithms can learn, as the number of steps needed to reach the target in each environment is reduced as the number of training episodes grows. Meanwhile, in a non-unique environment, only the algorithm proposed by this research can converge. The algorithm proposed by another research cannot converge, even at most episodes cannot reach the target after the episode ends, as shown by the line is not continuous at rooms 1 and 2. Meanwhile, in room 3, the agent cannot reach the target even once, shown by no line at all representing the data of another research method in a non-unique environment. This implies that if the assumption in research [8], [9] doesn't hold, the algorithm proposed by other research will be negatively impacted. Besides that, it also implies the algorithm proposed by other research is not robust under RF wave propagation characteristics.

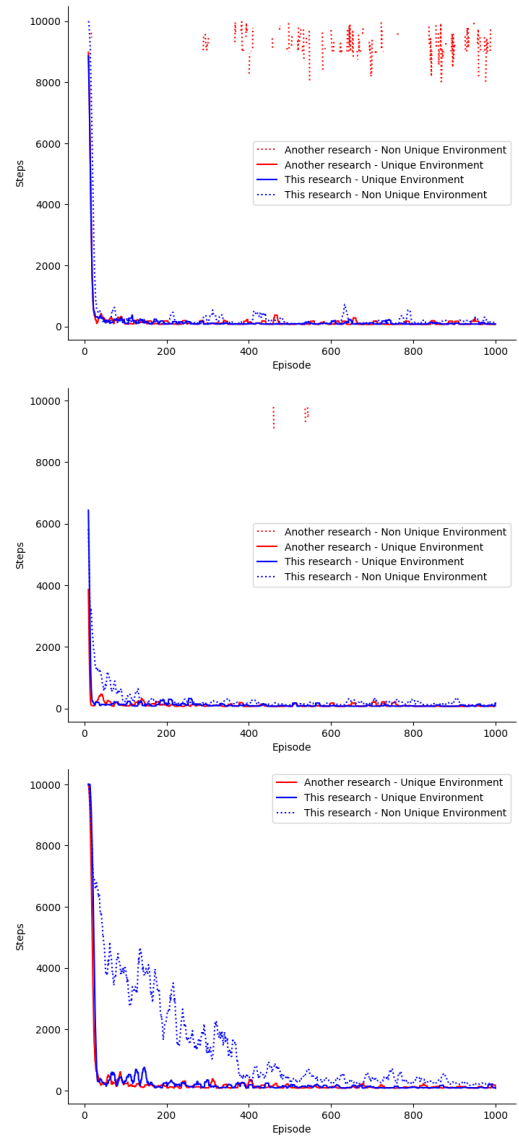


Fig. 7. Comparison of the number of steps needed by the agent to reach the target in Room 1 (above), Room 2 (middle), and Room 3 (bottom).

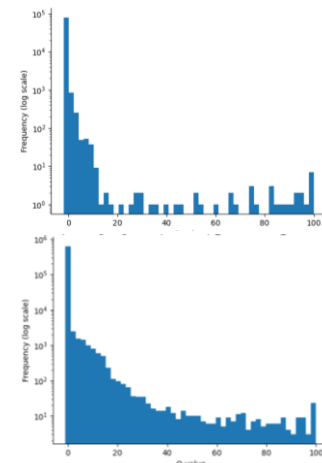


Fig. 8. The Q-values distribution of another research's algorithm (above) and proposed algorithm (below) in room 1 non-unique environment.

Fig. 8 shows the distribution of Q-values and how often Q-values appear after agents go through the learning process in room 1 with a non-unique environment. The graphic tells us that the more often the agent moves from one state to another, the more varied the Q-values that appear. It can be seen the algorithm proposed in this research has Q-value much more varied than the previous research algorithm, proving that this research algorithm able to increase the $P_a(s, s')$ from equation 2.

VI. CONCLUSION

Our research examined the approach used in previous studies for RSS-based target localization. These studies employed a single-variable Q-learning state using only RSS, which challenged the algorithm's convergence due to the uniform propagation of RF waves in all directions. Consequently, the RSS values were identical for locations equidistant from the target, making it difficult for the agent to determine its actual relative position. To overcome this limitation, our paper proposes the incorporation of multiple variables to define the Q-learning state. The proposed algorithm surpasses the single-variable approach's performance in reducing the number of steps required for the UAV to reach its target from the starting position. Additionally, we examined the assumption made by some existing research when creating simulation environments where no two locations have the same RSS value. In such environments, our proposed algorithm performs comparably to the Q-learning algorithm utilizing a single-variable Q-learning state.

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